

ITAMAR AMSALEM

Tenafly, NJ | 201-218-9050 | amsalemitamar@gmail.com | linkedin.com/in/itamarams/
github.com/itamaramsaleem

EDUCATION

Rutgers University

B.S. in Mathematics, B.A. in Computer Science

New Brunswick, New Jersey

September 2024 – May 2028

- **GPA:** 3.61 / 4.00 Dean's List (All Semesters)
- **Selected Coursework:** Mathematical Theory of Probability, Statistics, Linear Algebra, Multivariable Calculus, Differential Equations, Linear Optimization, Data Structures, Computer Architecture

PROFESSIONAL EXPERIENCE

Aspire Healthcare

Software Development Intern

Lakewood, New Jersey

July 2025 – September 2025

- Engineered Python ETL pipelines to extract, clean, and consolidate payroll and HPPD data across 200+ facilities and 10,000+ employees, replacing manual Excel workflows
- Built an automated HPPD analyzer comparing staffing efficiency metrics across facilities, reducing review time from 3 hours to 30 minutes
- Developed agentic ML pipelines on time-series staffing data to forecast labor costs and automate data-driven budget planning; integrated predictive models into ETL workflows

PROJECTS

GSD.AI

AI-Powered Restaurant Survival & Market Intelligence Platform

Rutgers BITS x IBM Datathon — 1st Place

February 2026

- Trained XGBoost binary classifier on ~40 day-zero features (ZIP market context, concept attributes, 23 cuisine flags) with no-leakage evaluation, class-imbalance weighting, and SHAP TreeExplainer; 91.9% accuracy, 0.82 ROC-AUC, 5-fold CV
- Computed per-ZIP Opportunity Scores via 20km Haversine spillover analysis — weighting cuisine gap (0.6×), log-scaled market size, and capped attribute bonuses — across 91 NJ zip codes from 163K+ Yelp reviews
- Deployed 7-stage FastAPI + React pipeline with 6-component Match Score algorithm (gap percentile, market stability, XGBoost survival bonus, hard risk penalties) ranking concepts across all NJ zip codes; built end-to-end in 24 hours, placing 1st

Trade the Knight

Algorithmic Trading Competition

Rutgers QFC Algorithmic Trading Competition — 1st Place

April 2026

- Developed and deployed two concurrent stat-arb bots using pairs trading strategies, trading live against a proprietary Rust-based exchange with a custom matching engine, realistic latency, and live order books
- Achieved ~100% return on \$1M notional with a 2.26 Sharpe ratio over 8 hours of live trading; edge driven by disciplined position sizing, risk controls, and multi-strategy coordination
- Placed 1st out of 80 competing traders; competition sponsored by Citadel, Optiver, and ThetaData

S&P 500 Return Predictor

Machine Learning Trading Strategy

Personal Project

January 2026

- Applied supervised learning (Random Forest, Logistic Regression) to predict S&P 500 return direction; implemented walk-forward backtesting framework to prevent lookahead bias
- Built features capturing price momentum and recent market trends across multiple time horizons (2 to 1000 days), and tuned a probability threshold to improve prediction precision
- Benchmarked models via precision, AUC-ROC, and out-of-sample Sharpe ratio to evaluate predictive robustness

TECHNICAL SKILLS

- *Quantitative Skills:* Regression Analysis, Classification Models, Time-Series Analysis, Feature Engineering, Backtesting, Cross-Validation, Data Wrangling, Risk Modeling, AI/ML Pipelines
- *Languages:* C, Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, LLM APIs), Java
- *Tools:* Excel, Jupyter Notebook, Git, Flask, FastAPI, Docker
- *Other:* Hebrew (fluent)

ADDITIONAL INFORMATION

Campus Involvement: Rutgers Quant Finance Club, Academic Chair at Alpha Epsilon Pi, Rutgers Poker Club

Interests: Alpine Skiing, Chess, Personal Finance, Poker, NBA Basketball (Knicks)